1D SDC represented as a Layered DAG

Jerry Kiely

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Questions

How do we confirm that the *minimum free energy* calculation arise from the *target configuration*?

Is the *target configuration* always the most *thermodynamically favoured configuration*?

MFE and Thermodynamic Favourability

The *dynamic programming* approach to calculating the *minimum free energy* of a 1D SDC

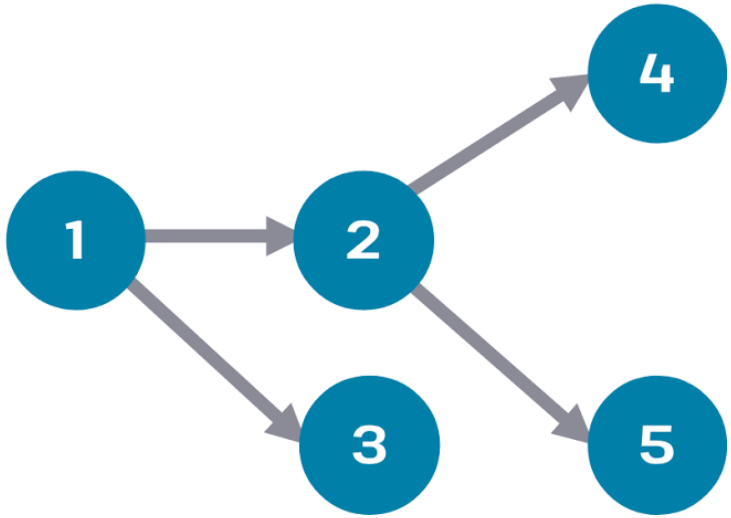
- N positions
- $\leq k$ tiles per position
- $\mathcal{O}(Nk^2)$

How do we know the calculated value of the *minimum free energy* is from the *target configuration*?

DAG

A Directed Acyclic Graph

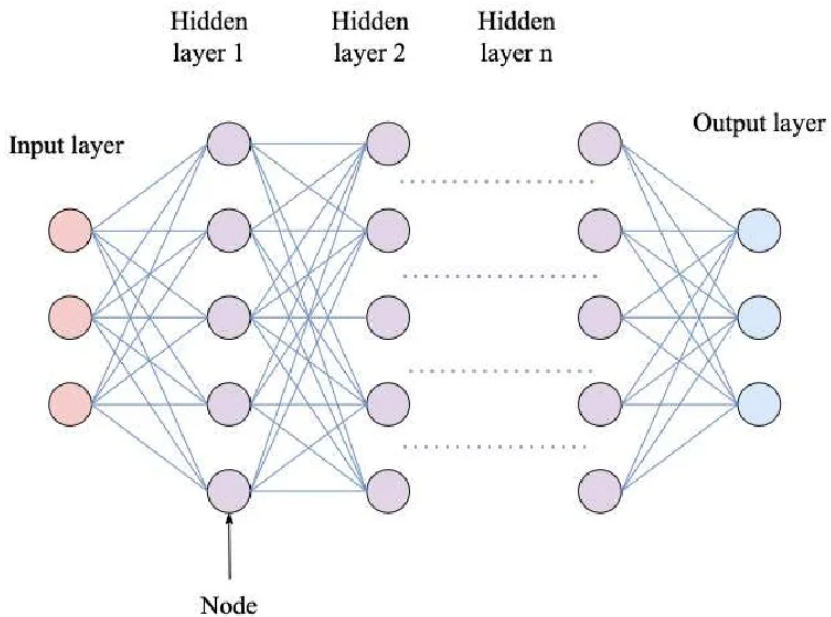
- Nodes
- Directed Edges
- No Cycles



Layered DAG

A Layered DAG

- Directed Acyclic Graph
- Partitioned into Layers
- Topological Ordering



Features of a Layered DAG

Why is a Layered DAG relevant to 1D SDC?



- Competing tiles occupy Layers (L_i)
- Layers are ordered ($L_i < L_j$ when $i < j$)
- A tile from L_i binds with a tile from L_{i+1}
- Edges connect tiles from layer i to layer $i + 1$
- Edges connect tiles with weight $\Delta G(T, r_i^m, l_{i+1}^n)$

MFE and Thermodynamic Favourability

For a Tileset represented as a Layered DAG, the Shortest Path gives you the most *thermodynamically favoured configuration*. If this is the *target configuration*, then the Shortest Path gives you:

- The *free energy* of the *target configuration*
- The *minimum free energy* of the Tileset

Traversal

The Shortest Path through a Layered DAG

- Topological Sort
- Edge Relaxation

```
for l in range(len(self.tiles) - 1):
    for u in self.get_layer(l):
        for v in self.get_layer(l + 1):
            uv = self.get_edge_weight(u, v, T)
            # relax edge u -> v
            if dist[v] > dist[u] + uv:
                dist[v] = dist[u] + uv
                pred[v] = u
```

Note: Layers induce a *topological ordering*, so a *topological sort* is unnecessary.

Comparison

Comparison of approaches:

	Time Complexity	Space Complexity
DP	$\mathcal{O}(Nk^2)$	$\mathcal{O}(k)$
DAG	$\mathcal{O}(V+E)$	$\mathcal{O}(V+E)$
Layered DAG	$\mathcal{O}(E)$	$\mathcal{O}(V+E)$

Note: For the Layered DAG, with respect to space complexity, we need E space to hold the edges, and V space to hold the reconstructed path itself.

Edges

If there are k_i competing tiles per layer L_i, then $k = \max_i(k_i)$

- k tiles per position
- k^2 possible bindings between tiles in consecutive positions
- N positions
- binding between $N - 1$ positions
- $(N - 1)k^2$ possible bindings between tiles in all positions
- $\therefore$ no more than $(N - 1)k^2$ edges
- $\therefore \mathcal{O}(E) = \mathcal{O}(Nk^2)$

Finally

Next steps?

- Partition Function
- Multilayer Perceptron (MLP)
- Free Palestine!

Goodbye

Questions?